

QQ10 Redox

WJEC TOPIC 3.1 & 3.2

IGC HK Exam



0	1
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Phosphorus burns in air to form phosphorus(V) oxide.
Give an equation for this reaction.

[1 mark]

01	$4\text{P} + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$ OR $\text{P}_4 + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$
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0 2

A student does an experiment to determine the percentage by mass of sodium chlorate(I), NaClO, in a sample of bleach solution.

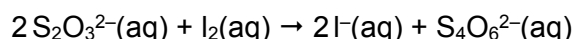
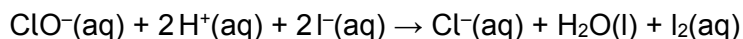
Method:

- Dilute a 10.0 cm³ sample of bleach solution to 100 cm³ with distilled water.
- Transfer 25.0 cm³ of the diluted bleach solution to a conical flask and acidify using sulfuric acid.
- Add excess potassium iodide to the conical flask to form a brown solution containing I₂(aq).
- Add 0.100 mol dm⁻³ sodium thiosulfate solution (Na₂S₂O₃) to the conical flask from a burette until the brown solution containing I₂(aq) becomes a colourless solution containing I⁻(aq).

The student uses 33.50 cm³ of sodium thiosulfate solution.

The density of the original bleach solution is 1.20 g cm⁻³

The equations for the reactions in this experiment are



0 2 . 1

Use all the information given to calculate the percentage by mass of NaClO in the original bleach solution.

Give your answer to 3 significant figures.

[7 marks]

Percentage by mass _____

02.1	M1 $n(\text{S}_2\text{O}_3^{2-}) = 33.50 \times 0.100 \div 1000 = \underline{0.00335}$ M2 $n(\text{I}_2) = 0.00335 \div 2 = 0.001675$ (from eqn 2) M3 $n(\text{ClO}^-)$ in 25 cm³ pipette = 0.001675 (from eqn 1) M4 $n(\text{ClO}^-)$ in 100 cm³ flask = 0.001675 $\times 4 = 0.00670$ = $n(\text{NaClO})$ in original 10cm³ sample M5 mass (NaClO) = 0.00670 $\times 74.5 = 0.499$ g M6 mass (bleach) = 10.0 $\times 1.20 = \underline{12}$ g M7 % by mass of NaClO = $\frac{0.499}{12} \times 100 = 4.16 \%$
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0 3

This question is about a glucose–oxygen fuel cell.

When the cell operates, the glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) molecules react with water at the negative electrode to form carbon dioxide and hydrogen ions.

Oxygen gas reacts with hydrogen ions to form water at the positive electrode.

0 3 . 1

Deduce the half-equation for the reaction at the negative electrode.

[1 mark]

0 3 . 2

Deduce the half-equation for the reaction at the positive electrode.

[1 mark]

0 3 . 3

Give the equation for the overall reaction that occurs in the Glucose–oxygen fuel cell.

[1 mark]

0 3 . 4

The negative electrode is made of carbon and the positive electrode is made of platinum.

Give the conventional representation for the glucose–oxygen fuel cell.

[2 marks]

0 3 . 5

State what must be done to maintain the EMF of this fuel cell when in use.

[1 mark]

Question	Answers	Additional comments/Guidelines	Mark
03.1	$\text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{H}_2\text{O} \rightarrow 6 \text{CO}_2 + 24 \text{H}^+ + 24 \text{e}^-$	Accept multiples	1
03.2	$\text{O}_2 + 4 \text{H}^+ + 4 \text{e}^- \rightarrow 2 \text{H}_2\text{O}$	Accept multiples	1
03.3	$\text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{O}_2 \rightarrow 6 \text{CO}_2 + 6 \text{H}_2\text{O}$	Accept multiples	1
03.4	$\text{C(s)} \mid \text{C}_6\text{H}_{12}\text{O}_6(\text{aq}), \text{H}^+(\text{aq}) \mid \text{CO}_2(\text{g}) \parallel \text{O}_2(\text{g}) \mid \text{H}^+(\text{aq}), \text{H}_2\text{O(l)} \mid \text{Pt(s)}$ OR $\text{C} \mid \text{C}_6\text{H}_{12}\text{O}_6, \text{H}^+ \mid \text{CO}_2 \parallel \text{O}_2 \mid \text{H}^+, \text{H}_2\text{O} \mid \text{Pt}$	M1 Must see following in correct order: $\text{C} \mid \text{C}_6\text{H}_{12}\text{O}_6 \text{ CO}_2 \parallel \text{O}_2 \text{ H}_2\text{O} \mid \text{Pt}$ M2 Cell completely correct Ignore H_2O on LHS Ignore state symbols Allow $\text{H}^+(\text{aq}) \mid \text{H}_2\text{O(l)}$ on RHS 0 marks if electrons included.	2
03.5	(Constantly) add reactants/glucose (and oxygen) OR keep concentration of reactants constant		1

0 4

This question is about electrode potentials and electrochemical cells.

0 4 . 1

State the meaning of the term electrochemical series.

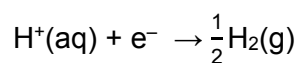
[1 mark]

Table 8 shows some electrode potentials.

Table 8

	$E^\ominus / \text{V}$
$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s}) + 6\text{H}_2\text{O}(\text{l})$	-0.44
$\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2(\text{g})$	0.00
$[\text{Co}(\text{NH}_3)_6]^{3+}(\text{aq}) + \text{e}^- \rightarrow [\text{Co}(\text{NH}_3)_6]^{2+}(\text{aq})$	+0.11
$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}(\text{aq}) + \text{e}^- \rightarrow [\text{Fe}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$	+0.77
$\text{VO}_2^+(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{VO}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+1.00
$[\text{Co}(\text{H}_2\text{O})_6]^{3+}(\text{aq}) + \text{e}^- \rightarrow [\text{Co}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$	+1.81

0 4 . 2

State **two** conditions needed for the following half-cell to have $E^\ominus = 0.00 \text{ V}$ 

[1 mark]

0 4 . 3

Identify the weakest reducing agent in **Table 8**.

[1 mark]



0 4 . 4

Use half-equations from **Table 8** to deduce an equation for the reduction of VO_2^+ to form VO^{2+} in aqueous solution by iron.

[2 marks]

0 4 . 5

Use data from **Table 8** to explain why $[\text{Co}(\text{H}_2\text{O})_6]^{3+}(\text{aq})$ will undergo a redox reaction with $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$

Give an equation for this reaction.

[2 marks]

Explanation

Equation

0 4 . 6

Suggest why the **two** cobalt(III) complex ions in **Table 8** have different electrode potentials.

[1 mark]

Question	Answers	Additional comments/Guidelines	Mark
04.1	(List of) electrode potentials/ $E^\ominus$ in (numerical) order OR half cells/equations in (numerical) order of electrode potential/ $E^\ominus$	Do not allow EMF in order	1

Question	Answers	Additional comments/Guidelines	Mark
04.2	Any 2 from 298 K or 25 °C [H ⁺] = 1 mol dm ⁻³ 100 kPa	Ignore 1 atm	1

Question	Answers	Additional comments/Guidelines	Mark
04.3	[Co(H ₂ O) ₆] ²⁺	Do not penalise absence of brackets	1

Question	Answers	Additional comments/Guidelines	Mark
04.4	$3 \text{VO}_2^+ + 6 \text{H}^+ + \text{Fe} + 3 \text{H}_2\text{O} \rightarrow 3 \text{VO}^{2+} + [\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ or $3 \text{VO}_2^+ + 6 \text{H}^+ + \text{Fe} \rightarrow 3 \text{VO}^{2+} + 3 \text{H}_2\text{O} + \text{Fe}^{3+}$ 1 mark for Fe ³⁺ as product and one mark for equation.	Ignore state symbols Allow 1 mark for balanced equation that gives Fe ²⁺ as product $2\text{VO}_2^+ + 4\text{H}^+ + \text{Fe} + 4\text{H}_2\text{O} \rightarrow 2\text{VO}^{2+} + [\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ or $2\text{VO}_2^+ + 4\text{H}^+ + \text{Fe} \rightarrow 2\text{VO}^{2+} + \text{Fe}^{2+} + 2\text{H}_2\text{O}$	2

Question	Answers	Additional comments/Guidelines	Mark
04.5	$E^\ominus \text{Co}^{3+}/\text{Co}^{2+} > \text{Fe}^{3+}/\text{Fe}^{2+}$	Allow electrode potential for Co^{3+} greater than for Fe^{3+} OR $1.81 > 0.77$ / EMF cell = 1.04 V	1
	$[\text{Co}(\text{H}_2\text{O})_6]^{3+} + [\text{Fe}(\text{H}_2\text{O})_6]^{2+} \rightarrow [\text{Co}(\text{H}_2\text{O})_6]^{2+} + [\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	Insist of reference to $E^\ominus$ in M1	1

Question	Answers	Additional comments/Guidelines	Mark
04.6	Different ligands	Penalise different concentrations/oxidation states	1

0 5

This question is about the development of lithium cells.
The value of $E^\ominus$ for lithium suggests that a lithium cell could have a large EMF.

Table 9 shows some electrode potential data.

Table 9

	$E^\ominus / \text{V}$
$\text{Li}^+(\text{aq}) + \text{e}^- \rightarrow \text{Li}(\text{s})$	-3.04
$2 \text{H}_2\text{O}(\text{l}) + 2 \text{e}^- \rightarrow \text{H}_2(\text{g}) + 2 \text{OH}^-(\text{aq})$	-0.83
$\frac{1}{2} \text{I}_2(\text{s}) + \text{e}^- \rightarrow \text{I}^-(\text{aq})$	+0.54

0 5**1**

Use data in **Table 9** to explain why an aqueous electrolyte is **not** used for a lithium cell.

[2 marks]

0 5**2**

In the 1970s lithium-iodine cells became a common power source for heart pacemakers. Lithium iodide is the final product of the cell reaction.

Use the data in **Table 9** to calculate the cell EMF of a standard lithium-iodine cell.

[1 mark]

0 5**3**

An EMF value for a commercial lithium-iodine cell is 2.80 V

Suggest why this value is different from the value calculated in Question **09.2**.

[1 mark]

0 5 . 4

In some lithium cells, lithium perchlorate (LiClO_4) is used as the electrolyte.

Deduce the oxidation state of chlorine in LiClO_4

[1 mark]

In other lithium cells, lithium cobalt oxide electrodes **and** lithium electrodes are used.

0 5 . 5

Give an equation for the reaction that occurs at the positive lithium cobalt oxide electrode.

[1 mark]

0 5 . 6

Give an equation for the reaction that occurs at the negative lithium electrode.

[1 mark]

7

Question	Answers	Additional comments/Guidelines	Mark
05.1	Lithium would react with the electrolyte/water $E^\ominus$ for Li^+/Li more negative than for water or $\text{EMF} = 2.21(\text{V})$ or $E^\ominus \text{Li}^+/\text{Li} < \text{H}_2\text{O}/(\text{H}_2, \text{OH}^-)$	Allow water will oxidise Li to Li^+ or Li will reduce water to hydrogen	1
		Ignore EMF is negative	1

Question	Answers	Additional comments/Guidelines	Mark
05.2	$0.54 - (-3.04) = \underline{3.58} (\text{V})$		1

Question	Answers	Additional comments/Guidelines	Mark
05.3	Non-standard conditions	Allow non-aqueous conditions or different conditions	1

Question	Answers	Additional comments/Guidelines	Mark
05.4	(+) 7	Accept VII	1

Question	Answers	Additional comments/Guidelines	Mark
05.5	$\text{Li}^+ + \text{CoO}_2 + \text{e}^- \rightarrow \text{Li}^+\text{CoO}_2^-$ or $\text{Li}^+ + \text{CoO}_2 + \text{e}^- \rightarrow \text{LiCoO}_2$		1

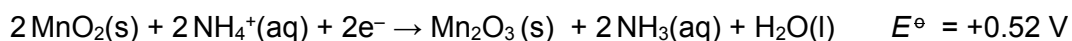
Question	Answers	Additional comments/Guidelines	Mark
05.6	$\text{Li} \rightarrow \text{Li}^+ + \text{e}^-$		1

0 6

This question is about cells.

0 6 . 1

The half-equations for two electrodes that combine to make a non-rechargeable cell are

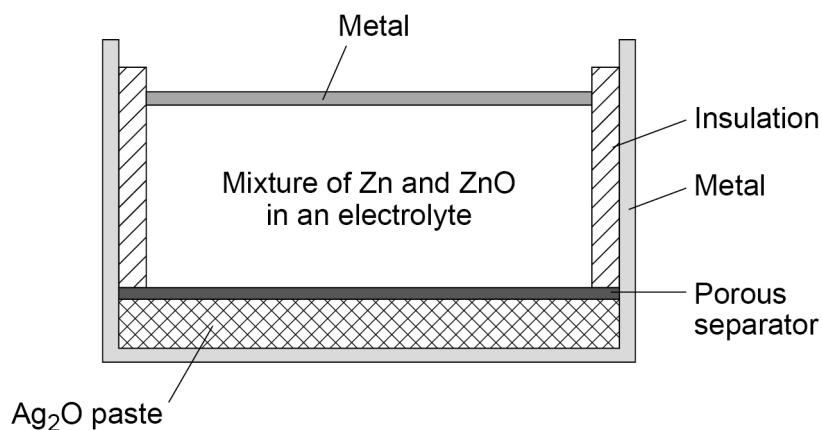


Identify the oxidising agent in this cell.

[1 mark]

Figure 1 shows a cross-section through a rechargeable silver–zinc cell.

Figure 1



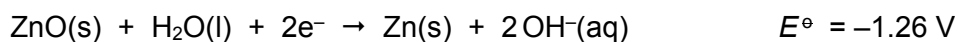
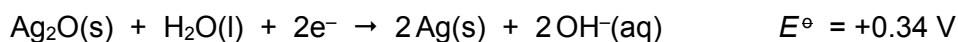
0 6 . 2

Suggest the function of the porous separator in **Figure 1**.

[1 mark]

0 6 . 3

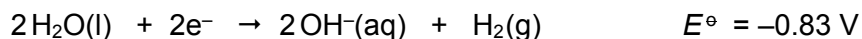
The standard electrode potentials for two half-equations for the silver–zinc cell are



Give an equation for the overall reaction that occurs when the cell is recharging.

[1 mark]

The EMF of an alkaline hydrogen–oxygen fuel cell is +1.23 V
The standard electrode potential for one of the electrodes in the
alkaline hydrogen–oxygen fuel cell is



0 6 . 4

Give the half-equation for the other electrode and calculate its standard electrode potential.

[2 marks]

Equation

$E^\ominus$

0 6 . 5

Suggest why the EMF values of the acidic and alkaline hydrogen–oxygen fuel cells are the same.

[1 mark]

6

END OF QUESTIONS



Question	Answers	Additional Comments/Guidelines	Mark
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06.1	MnO ₂		1 AO2
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Question	Answers	Additional Comments/Guidelines	Mark
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06.2	allows ions to move/flow/transfer or to complete the circuit or acts as a salt bridge	ignore to allow current/charge to flow do not accept electrons to flow	1 AO1
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Question	Answers	Additional Comments/Guidelines	Mark
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06.3	$2\text{Ag} + \text{ZnO} \rightarrow \text{Zn} + \text{Ag}_2\text{O}$	ignore state symbols	1 AO3
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Question	Answers	Additional Comments/Guidelines	Mark
06.4	$\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^- \rightarrow 4\text{OH}^-(\text{aq})$ $E^\ominus = (+)0.4(0) \text{ (V)}$	ignore state symbols allow multiples	1 1 AO1 AO2

Question	Answers	Additional Comments/Guidelines	Mark
06.5	same <u>overall</u> reaction or $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$		1 AO2
